

Panoptic Flavor Tagging and Jet Calibration

DFM-NOTE-2026-03

UTA Detector Foundation Model group

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August 2026 • github.com/afarbin/DetectorFoundationModel, branch dfm-merge**Abstract**

Following the finding of DFM-NOTE-2026-02 that flavor-conditioned calibration is optimal but presumes flavor knowledge, we train a single panoptic head emitting flavor probabilities and calibration jointly: $[p_b, p_c, p_{\text{light}}, \mu, \sigma]$ with a combined cross-entropy + Gaussian NLL objective, on the TJC-graph backbone of DFM-NOTE-2026-01 and the 1.2M-jet all-flavor dataset. Joint training costs tagging nothing—AUC 0.941 vs 0.939 for an identically-sized dedicated tagger, with light-jet rejection 117 ± 5 at the 70% b -efficiency working point and consistently better c -rejection—while calibration lands within $\sim 2\%$ relative of the truth-conditioned optimum (overall JER 0.126 vs 0.123), with charm improving under joint training. The panoptic model thus supplies its own conditioning signal, realizing the proposal’s joint-reconstruction claim at per-jet scale.

1 Introduction

DFM-NOTE-2026-02 [2] showed that the best calibration strategy conditions a shared model on jet flavor—information that at inference must itself be inferred. This note closes that loop: one head predicts both, trained jointly. Backbone, dataset, calibration metrics, and baselines are those of Refs. [1, 2]; tagging is evaluated with flavor-tagging conventions (b -efficiency working points at 70, 77, 85%; background rejection = $1/\text{efficiency}$), using p_b as the discriminant.

2 Architecture and training

The head emits five outputs: three flavor logits and $(\mu, \log \sigma^2)$. Loss: $\text{CE}(\text{flavor}) + \lambda \times \text{NLL}(\text{response})$, $\lambda = 1$; the $\lambda = 0$ limit trains the identical architecture as a pure tagger and serves as the tagging baseline. Three seeds each on the full all-flavor dataset.

3 Results

	AUC(b)	light rej @70%	c rej @70%	light @77%	light @85%
panoptic (joint)	0.941 ± 0.001	117 ± 5	7.2 ± 0.2	60 ± 2	21.0 ± 0.6
dedicated tagger	0.939 ± 0.001	122 ± 10	6.5 ± 0.3	58 ± 3	19.5 ± 0.9

Table 1: Tagging performance, isolated test jets, 3 seeds.

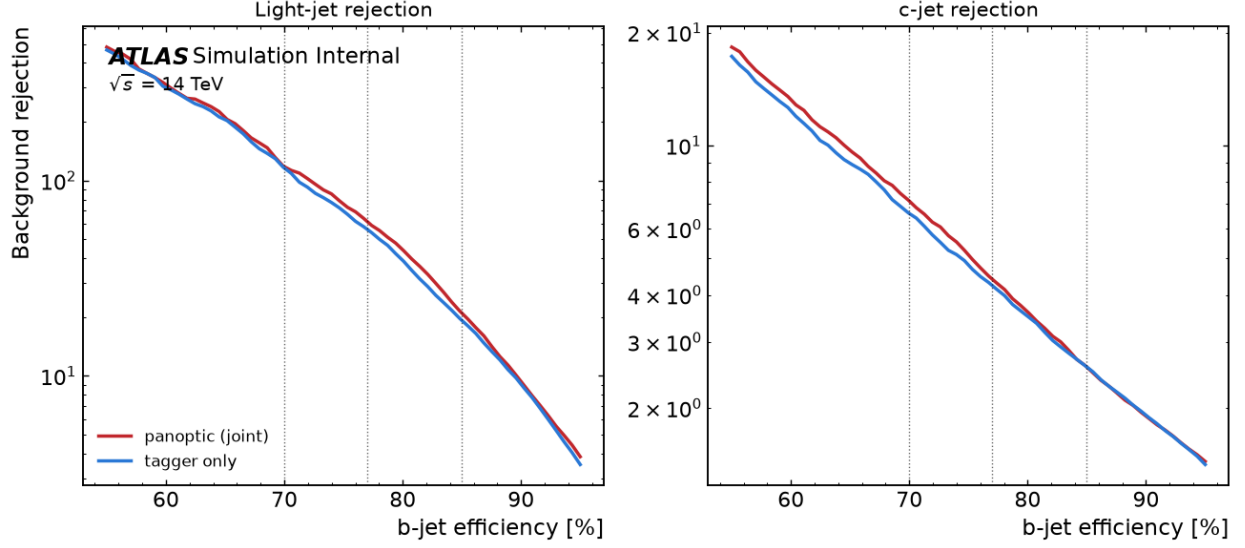


Figure 1: Background rejection versus b -efficiency for light (left) and c jets (right): the joint model (red) matches or exceeds the dedicated tagger (blue) across the working range; working points marked.

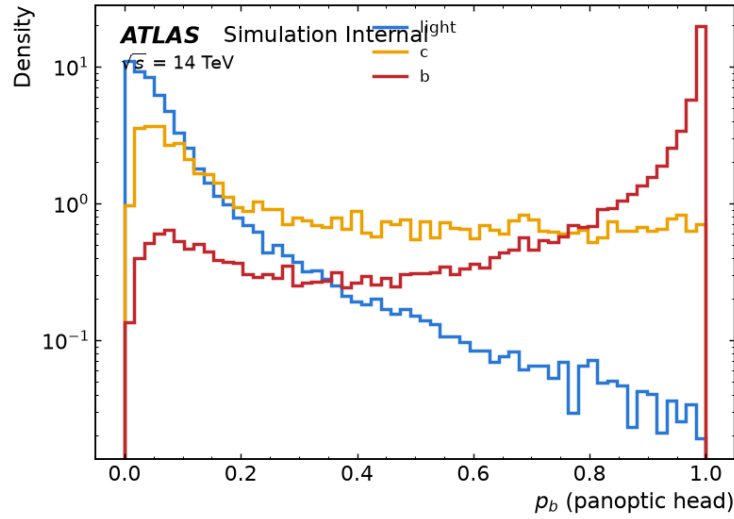


Figure 2: Panoptic p_b by true flavor: clean b /light separation with c intermediate, as physics dictates.

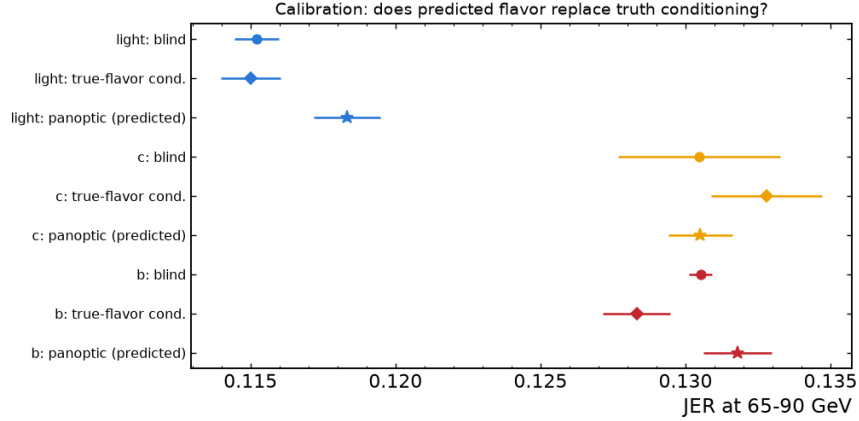


Figure 3: Per-flavor JER at 65–90 GeV: blind (circles) and truth-conditioned (diamonds) from Note 02, versus the panoptic model with predicted flavor (stars). The panoptic model recovers most of the conditioning benefit—and improves charm—without truth labels at inference.

Calibration: overall $\text{JER}_{@65-90} = 0.126$ versus 0.123 for the truth-conditioned model of Ref. [2] and 0.124 blind; closure RMS 4.3%. Per flavor (seed 0): light 0.120, c 0.129 (better than truth-conditioned 0.133), b 0.133 (vs 0.128). The dedicated-tagger run’s calibration outputs are untrained by construction and excluded from comparison.

4 Conclusions and outlook

A single panoptic head performs flavor tagging at parity-or-better with a dedicated tagger while calibrating within $\sim 2\%$ of the truth-conditioned optimum—the two tasks share representation, and each supplies what the other needs. Next steps in the program: event-level missing- E_T estimation (truth MET derivable from the neutrino record) and query-based jet finding, extending the panoptic principle from per-jet to per-event reconstruction.

References

- [1] DFM-NOTE-2026-01, *Machine-Learned Jet p_T Calibration from Low-Level Detector Information*, [notes/DFM-NOTE-2026-01-jet-pt-calibration/](#).
- [2] DFM-NOTE-2026-02, *Flavor-Dependent Machine-Learned Jet Calibration*, [notes/DFM-NOTE-2026-02-flavor-calibration/](#).